

Activation-space alignment to a behavior direction predicts a persona’s own base rate but does not beat the base prior on where leakage lands

Explore Persona Space, experiment record #657

Run date: 18 June 2026 | Model: Qwen2.5-7B-Instruct

Abstract

A prior result found that how strongly a persona “points toward” the sycophancy direction inside a language model predicts how sycophantic that persona is on its own (Spearman $\rho = 0.73$). We test two extensions of that geometric signal: whether the alignment $\rightarrow$ base-rate link generalizes to other behaviors, and whether the same geometry predicts *where an implanted behavior leaks* across held-out personas. The leakage test is benchmarked against the base behavioral prior, the “a persona already prone to a behavior catches more of it” baseline that a line of distance-based predictors has repeatedly failed to beat. Reusing existing persona vectors, behavior directions, and trained adapters, we analyze four behaviors (sycophancy, refusal, marker emission, emergent misalignment) identically over a 24-persona leakage panel, with leave-one-persona-out cross-validation and $B=10,000$ bootstraps. The base-rate link generalized to one new behavior, not three: alignment predicts a persona’s own sycophancy ($\rho=0.68$, $n=16$) and emergent misalignment ($\rho=0.51$, $n=23$) but is flat for refusal ($\rho=0.01$). On leakage, alignment failed out of sample for every behavior; only after partialling the prior twice does a sycophancy signal appear ($\rho=0.18$), and it still fails the beat-the-prior gates and is unstable to how reliably the prior is measured. A $17\times$ -larger pooled re-read (399 cells) tightens the verdict but does not flip it: pooled doubly-partialled $\rho=0.088$, 95% CI $[-0.017, 0.171]$. Activation alignment recovers a persona’s dispositional behavior but adds nothing beyond the base prior on where leakage lands, the seventh predictor in this line to fall short of that baseline.

1 Introduction

Fine-tuning a language model to install a behavior into one “source” persona rarely keeps the behavior confined: it *leaks* onto other personas and onto the default assistant context. A practical safety question follows: can we predict, before training, *which* held-out personas a given implant will spill onto? A natural candidate predictor is geometric. Recent work identifies linear directions in a model’s activation space that mediate character traits and behaviors: persona vectors for traits such as sycophancy and “evil” [3], a single refusal direction [1], and “misaligned persona” features that control emergent misalignment [4, 2]. If a persona’s activation-space *alignment* to a behavior’s direction (the cosine between its persona vector and that direction) says how much the behavior will land on it, that would give a cheap, training-free predictor of leakage.

The prior result and the open question. A precursor experiment (#623) found exactly such a signal for one behavior: the cosine alignment of a persona vector to the sycophancy direction predicts that persona’s *own* base sycophancy rate, Spearman $\rho=0.726$ ($n=35$, 95% bootstrap CI $[0.49, 0.87]$). That is a base-rate prediction, not a leakage prediction. Separately, a line of leakage-prediction experiments (#518, #545, #605, #603, #649) found that no geometric or distance-based predictor reliably beats a simple baseline: the *base behavioral prior*, i.e. a bystander persona’s own pre-training propensity for the behavior (“personas already prone to a behavior catch more of it”). Across that line, a unit’s own base prior keeps out-predicting geometry. The project’s leakage theory frames the

same tension: a predictor earns its keep only if it explains leakage variance *beyond* the base prior, not if it merely correlates with it.

What this experiment changes. This study (#657) changes one variable relative to its parent #623: the prediction *target* moves from a persona’s base rate to cross-persona leakage landing, benchmarked against the base prior. Concretely we test two hypotheses across four behaviors:

- **H1 (generalization).** Does the alignment $\rightarrow$ base-rate link hold for behaviors beyond sycophancy (refusal, marker emission, emergent misalignment)? Pre-registered bar: $\rho \geq 0.5$ with a positive-bounded CI on ≥ 3 of 4 behaviors.
- **H3 (beat the prior on leakage).** Does the same alignment geometry predict where an implant leaks across held-out personas, *beyond* the base behavioral prior? Pre-registered bar: out-of-sample (held-out) doubly-partialled ρ with a CI clearing zero, an incremental-fit (ΔR^2) CI clearing zero, and a reliability band that confirms on ≥ 1 primary behavior.

A behavior’s direction first has to pass a minimal causal sign-of-life gate (adding the direction must increase the judged behavior) before its predictive correlation is read; the beat-the-prior test runs only on directions that are pure base-model reads and pass that gate. The whole study is training-free reuse: the only new compute is one activation-extraction pass.

Contribution. The two targets dissociate. Activation alignment is a real read of a persona’s *dispositional* behavior, recovering the base rate for sycophancy and emergent misalignment. But it carries no information about *leakage landing* once the base prior is accounted for, on any behavior and at $17\times$ the sample size.

2 Methods

All measurements are on **Qwen2.5-7B-Instruct** (BF16). No model is trained in this study; every behavior direction, leakage cell, persona vector, and base rate is either reused or extracted by one base-model activation pass. Following a self-contained-method discipline, we state the full procedure that produced each reused artifact rather than deferring to the source experiment.

2.1 Persona vectors

A persona vector encodes “how the model represents acting as persona i ” as a centroid difference. For each of a panel of personas we build a chat prompt from that persona’s system prompt over a shared bank of 240 extraction questions, run a forward pass, and read the residual-stream hidden state at the last prompt token (headline readout; a response-averaged readout is a robustness variant). Averaging over the 240 questions gives a per-persona centroid; the persona vector is $v_i = \text{centroid}_i - \text{centroid}_{\text{assistant}}$ at each of layers $\{7, 14, 21, 27\}$. The bank is global-mean-centered. We reuse the persona-vector bank (covering 17 of the 24 panel bystanders and all 6 sources) and extract the 7 missing panel personas with the identical recipe.

2.2 Behavior directions

Each behavior’s direction is constructed by a behavior-appropriate recipe, because the generic trait recipe could not pass the causal gate for content-triggered behaviors.

- **Sycophancy (primary, base-model read).** Reused verbatim from the parent experiment’s sycophancy trait direction, built by the difference-in-means recipe of Chen et al. [3]: from a natural-language trait description, generate positive/negative instruction pairs and extraction questions, sample 10 rollouts per (question $\times$ pos/neg instruction) on the base model, judge each 0–100 for the trait, retain pos rollouts scored > 50 and neg rollouts scored < 50 , and set the

direction to $\text{mean}(\text{positive activations}) - \text{mean}(\text{negative activations})$ per layer (realized retain counts: 593 positive, 999 negative).

- **Refusal (primary, base-model read).** Difference-in-means in the style of Ardit et al. [1]: capture the post-instruction-token residual-stream activation over 128 harmful instructions (AdvBench behaviors, `mlabonne/harmful_behaviors`) and 128 harmless instructions (`mlabonne/harmless_alpaca`); the direction is $r_\ell = \text{mean}(\text{harmful}) - \text{mean}(\text{harmless})$ at each layer. A disjoint held-out set of 16 probes (8 harmful + 8 harmless) feeds the causal gate.
- **Emergent misalignment (secondary, trained-shift read).** Read as a trained–base activation shift off a reused LoRA adapter trained on a Betley-style bad-medical-advice corpus ($r=32$, $\alpha=256$, rsLoRA, lr $2e-5$, 375 steps, seed 42, full-response cross-entropy). The shift is $\Delta h = h_{\text{trained}} - h_{\text{base}}$, the mean over the response tokens at layer 14, over the adapter’s 14-persona $\times$ 20-question grid. Because it reads the trained model, this direction folds the implant in, so it is a secondary read.
- **Marker “*” (secondary / base-rate-only).** Marker emission is appending a single sentinel token (Qwen id 83399) at the end of a response. The intended affordance-contrastive direction failed the causal gate, so the run fell back to a trained-shift read off a marker-implant adapter ($r=8$, $\alpha=16$, rsLoRA, lr $2e-6$, 600 steps; marker-only loss with $\sim 1:1$ contrastive negatives), which demotes marker to a secondary read.

2.3 Leakage cells and the leakage Δ

A leakage cell is one (source persona, bystander persona) pair. Each source adapter is a LoRA fine-tune that installs the behavior into the source persona using the project’s contrastive recipe ($r=32$, $\alpha=64$ rsLoRA, lr $1e-5$, 3 epochs, seed 42; ~ 700 contrastive rows per source: source-positive rows plus bystander-negative and no-persona rows so the behavior is gated to the source). Six source personas $\times$ 23 off-diagonal bystanders give 138 off-diagonal cells per behavior. The leakage Δ for a cell is the trained-minus-base behavior rate on the bystander persona, measured on-policy (the model writes its own answer, scored at the natural slot; the marker Δ is the trained–base $\log P(*)$ shift at the end of the model’s own response). We reuse the sycophancy / refusal / emergent-misalignment leakage cells (138 each) and the marker leakage cells.

2.4 The alignment predictor and the bake-off

The predictor is a per-persona scalar $\text{align}_i = \cos(v_i, d_{\text{behavior}})$ at layer 14, where d_{behavior} is the behavior direction. For H1 we correlate align_i with persona i ’s base rate (raw Spearman ρ , bootstrap CI over personas). For H3 the dependent variable is the held-out (leave-one-persona-out) Spearman ρ between align and the leakage Δ landing on a bystander, with the base prior partialled out two ways: the scalar base rate, and a geometry-side restatement of it ($\cos(v_i, b_{\text{prior centroid}})$, where the prior centroid is the base-rate-weighted mean persona vector). We report the raw, singly-partialled, and doubly-partialled ρ , a grouped-cross-validation ΔR^2 for alignment-beyond-prior, and a 3-point reliability band that re-runs the doubly-partialled ρ under assumed levels of how reliably the base prior is measured. Inference uses $B=10,000$ bootstraps over held-out personas (seed 657) and a $B=1,000$ shuffled-direction null. A pooled secondary read (Section 3.4) raises sample size by pooling the three in-scope behaviors at the predictor-skill level.

2.5 Causal sign-of-life gate and judge

Each direction must show that *adding it increases the judged behavior* before its ρ is read. We add the direction at coefficients $\{4, 8, 12\}$ across layers $\{7, 14, 21, 27\}$, generate, and score each completion 0–100 for the target behavior with `claude-sonnet-4-5` as judge; a direction passes if the headline-layer effect is positive. The leakage / base-rate dependent variables are reused on-policy

Table 1: Complete configuration. Training is reuse-only; the only new compute is one activation-extraction pass on $1 \times$ A100-80.

Parameter	Value
Base model	Qwen2.5-7B-Instruct
Training (this study)	none (reuse-only analysis)
Persona vector	$\text{centroid}_{\text{persona}} - \text{centroid}_{\text{assistant}}$, global-mean-centered
Persona-vector readout	last-prompt-token (headline); response-avg (robustness)
Layers swept	$\{7, 14, 21, 27\}$; headline layer 14
Alignment metric	cosine
Refusal direction	Arditi diff-in-means, 128 harmful + 128 harmless, 16 held-out probes
EM / marker directions	trained-base activation shift @ L14 (fold the implant in)
EM adapter	$r=32$, $\alpha=256$, rsLoRA, lr $2e-5$, 375 steps, seed 42
Marker adapter	$r=8$, $\alpha=16$, rsLoRA, lr $2e-6$, 600 steps, marker-only loss
Leakage-source adapters	$r=32$, $\alpha=64$ rsLoRA, lr $1e-5$, 3 epochs, ~ 700 contrastive rows
Causal gate	add direction $\{4, 8, 12\} \times \{L7, 14, 21, 27\}$; pass if headline effect > 0
Judge (causal gate)	claude-sonnet-4-5
Judge (reused base rate)	claude-haiku-4-5 (sycophancy, inherited)
Partial correlation	Spearman, doubly-partialled on [base rate, geometry-side prior]
Bootstrap	$B=10,000$ over held-out personas, seed 657
Shuffled-direction null	$B=1,000$ random rotations, seed 657
Held-out protocol	leave-one-persona-out (primary); leave-one-behavior-out (pooled)
Effective N (leakage)	$\approx 17-23$ bystanders per behavior; $\text{MDE-}\rho \approx 0.62$ at $n=17$

judge measurements; the reused sycophancy base rate was judged by **claude-haiku-4-5** (inherited from its source pass), and the refusal/emergent-misalignment leakage rates by **claude-haiku-4-5** and **claude-sonnet-4-5** respectively. Table 1 gives the complete hyperparameter set.

3 Results

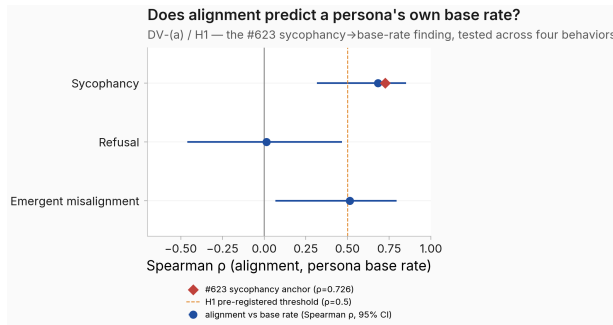
3.1 H1: alignment predicts a persona’s own base rate for sycophancy and EM, not refusal

Figure 1a plots, per behavior, the raw Spearman ρ between a persona’s alignment and that persona’s own base rate, with a 95% bootstrap CI over personas. The pre-registered generalization bar ($\rho \geq 0.5$ with a positive-bounded CI on ≥ 3 of 4 behaviors) held on two. Sycophancy reproduced the parent ($\rho=0.68$, CI $[0.32, 0.85]$, $n=16$; the parent’s 0.726 falls inside the CI, so the data join is correct). Emergent misalignment cleared by point estimate only ($\rho=0.51$, CI $[0.07, 0.79]$, $n=23$), its lower CI bound below 0.5. Refusal was flat ($\rho=0.01$, CI $[-0.46, 0.47]$, $n=23$). Marker is uninterpretable here ($n=3$) and omitted. The result is a partial generalization: alignment recovers dispositional behavior for the two trait-like behaviors and not for refusal.

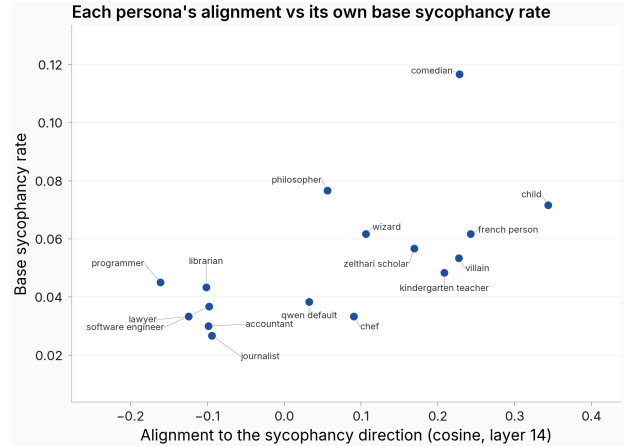
Figure 1b unpacks the sycophancy point into its 16 underlying personas (one dot each, no aggregation). The base rate climbs with alignment from the low-alignment cluster near rate 0.03 to higher-alignment personas near 0.06–0.12; the negative-alignment personas all sit at low base rates, so the rank correlation is not driven by a single outlier.

3.2 The causal gate is weak, and the marker direction fails it

Figure 2a shows each direction’s causal sign-of-life effect at its headline layer, in judge points on the 0–100 scale. Every effect is $\ll 1\%$ of scale: refusal $+0.96$ (the mean of per-strength effects $[1.75, -2.75, 3.875]$, i.e. non-monotonic), emergent misalignment $+0.07$ (positive only at the strongest strength), and marker exactly 0.0 at all 12 cells. The marker affordance direction



(a) Per behavior: Spearman $\rho(\text{alignment, base rate})$, 95% bootstrap CI. Sycophancy 0.68 and EM 0.51 clear the dashed $\rho=0.5$ bar; refusal near 0 with a wide CI. The diamond marks the parent sycophancy anchor (0.726).



(b) The 16 personas behind the sycophancy point. $x = \cos(\text{persona vector, sycophancy direction})$ at L14; $y = \text{base sycophancy rate}$. The upward trend is what $\rho=0.68$ compresses; no single persona drives it.

Figure 1: H1 (base rate). Alignment predicts a persona’s own rate for sycophancy and EM, not refusal (2 of 4).

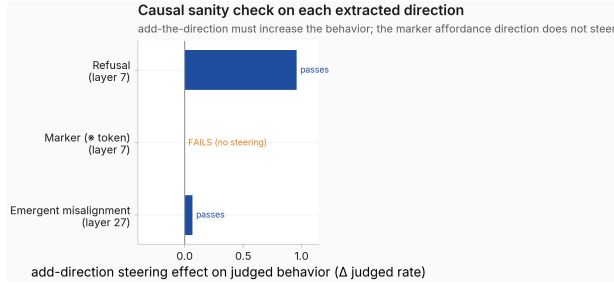
does not move marker emission at all, so it does not measure the behavior; the run fell back to the trained-shift read. Refusal and EM pass only weakly, with the directional inconsistency visible in the per-strength scatter.

Figure 2b exposes a sharper problem: the gate passes at layer 7 (refusal) and layer 27 (EM), but the alignment cosine driving the predictive ρ is read at layer 14. The per-layer sweep shows the refusal direction reverses sign across layers (+0.96 at L7, -1.31 at L14, -3.52 at L21, -4.67 at L27). At the L14 readout the refusal direction steers *negative*, away from the gate-validated direction. The refusal base-rate null (Section 3.1) may therefore be a wrong-layer artifact; re-reading at the gate-passing layer is a follow-up, not done here.

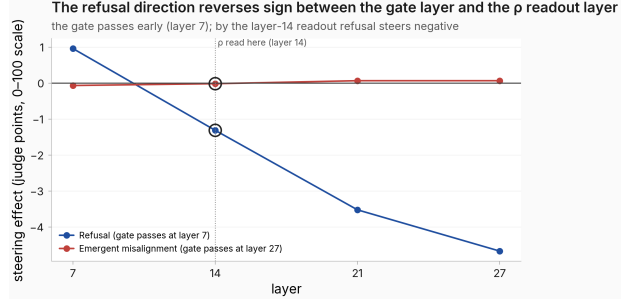
3.3 H3: alignment did not predict leakage out of sample, or beyond the base prior

Figure 3a shows the held-out alignment-vs-leakage correlation per behavior in three forms: raw (no controls), singly-partialled (vs the base prior only), and doubly-partialled (vs base prior + geometry-side prior), with 95% CIs. Out of sample, the raw held-out ρ crosses zero for *every* behavior (sycophancy raw $\rho=0.14$, CI $[-0.01, 0.30]$). A signal appears only after the prior is partialled out: the doubly-partialled sycophancy $\rho=0.18$ (CI $[0.05, 0.33]$) is the only interval clearing zero. Because the raw out-of-sample read is null, that clearance is a partialling artifact, not an out-of-sample prediction. Refusal (doubly-partialled $\rho=0.12$, CI $[-0.04, 0.25]$) and EM ($\rho=-0.07$, CI $[-0.24, 0.16]$) straddle zero; EM falls below its shuffle null ($p=0.89$). Even the lone sycophancy clearance fails the beat-the-prior gates: its incremental-fit ΔR^2 CI spans zero ($[-0.060, 0.133]$) and its reliability-band verdict is INDETERMINATE.

Figure 3b makes the instability explicit. The beat-the-prior test partials out the base prior, and how reliably the prior is measured changes how much it removes. The sycophancy doubly-partialled ρ is 0.19 at the observed reliability and inflates toward 1.0 only as the prior is assumed measured-with-error. Clearing only at that optimistic end is the signature of an inconclusive signal, not a stable beat. Refusal is flat (~ 0.11) across the whole band, consistent with no signal under any reliability assumption.

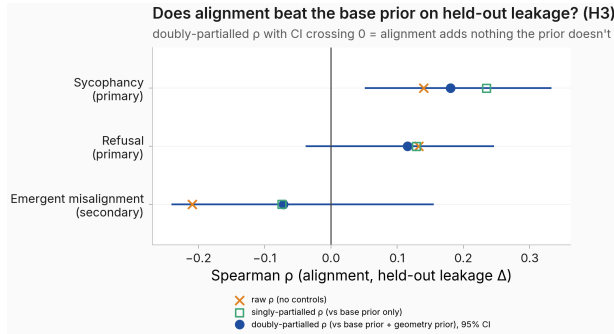


(a) Headline-layer causal effect (judge points, full scale 0–100); diamond = mean, dots = the three per-strength effects. Refusal +0.96, EM +0.07, marker 0 everywhere.

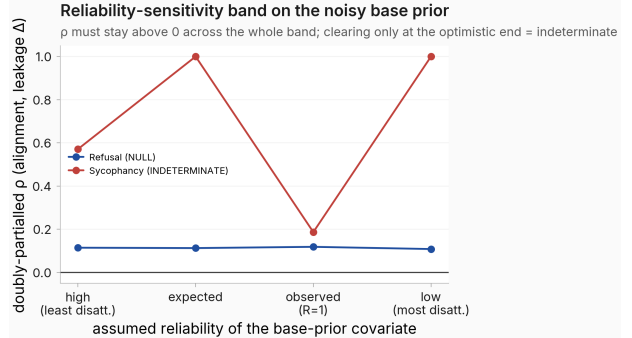


(b) Per-layer causal effect. The refusal direction reverses sign between its L7 gate and the L14 ρ readout (circled); EM stays flat near zero.

Figure 2: Causal sign-of-life gate. Each gate effect is $\ll 1\%$ of scale, and the refusal direction sign-flips between its gate layer and the readout layer.



(a) Held-out $\rho(\text{alignment, leakage } \Delta)$: raw ($\times$), singly-partialled ($\square$), doubly-partialled ($\bullet$, 95% CI). Only the doubly-partialled sycophancy interval clears zero; the raw read is null for all.



(b) Doubly-partialled ρ across assumed prior-reliability. Sycophancy is 0.19 at the observed point and inflates to 1.0 only at the assumed-mismeasured end; refusal is flat near 0.11.

Figure 3: H3 (beat the prior on leakage). Out-of-sample prediction is null; the lone partialled-sycophancy clearance is an artifact and is unstable to the prior’s assumed reliability.

3.4 A 17 $\times$ -larger pooled re-read tightens the verdict but does not flip it

The per-behavior reads are power-limited at ≈ 23 held-out personas (minimum detectable $\rho \approx 0.62$ at $n=17$), so a low-power null could masquerade as a true null. A higher- N secondary read pools the three in-scope behaviors (marker excluded) at the predictor-skill level, lifting the sample to 399 cells across 69 (behavior, bystander) groups. Figure 4 compares the three per-behavior reads against the pooled read. The pooled doubly-partialled $\rho=0.088$ (CI $[-0.017, 0.171]$) still straddles zero (no beat, no confirmed null), and the grouped-CV ΔR^2 point is 0.002 with an interval crossing zero. Supporting reads agree: grouped-CV held-out $\rho=0.020$ and strict leave-one-whole-behavior-out $\rho=0.116$. At 17 $\times$ the sample the interval tightens but the verdict does not flip; the per-behavior null was not a low-power artifact. Table 2 collects the headline numbers.

Why the marker behavior was untestable. Three independent exclusions each suffice. (i) The affordance-recipe marker direction had zero steering at all 4 layers $\times$ 3 strengths (causal gate failed). (ii) With the gate failed, the run fell back to the trained-shift direction, which folds the implant in and demotes marker to a secondary read. (iii) The marker leakage panel is context-keyed and overlaps the 24-persona axis on only 4 personas (`assistant`, `comedian`, `software_engineer`,

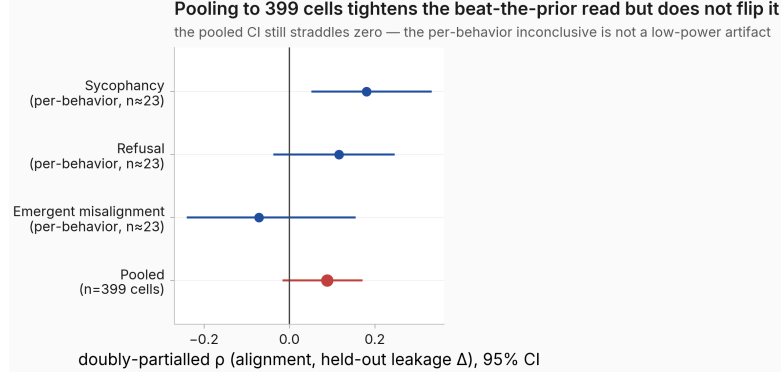


Figure 4: Pooled higher- N read. Doubly-partialled $\rho(\text{alignment, held-out leakage } \Delta)$, 95% CI: the three per-behavior reads ($n \approx 23$ each) above the pooled read ($n=399$). Pooling tightens the interval but it still straddles zero (pooled $\rho=0.088$, CI $[-0.017, 0.171]$).

Table 2: Headline numbers. H1 is the raw base-rate ρ ; H3 is the held-out doubly-partialled leakage ρ (DP) with its 95% CI. Marker is excluded from H3 (causal-gate failure + panel overlap $4 < 8$).

Behavior	H1 base-rate ρ	H3 raw ρ	H3 DP ρ	H3 DP CI	Verdict
Sycophancy	0.68 ($n=16$)	0.14	0.18	$[0.05, 0.33]$	inconclusive
Refusal	0.01 ($n=23$)	0.13	0.12	$[-0.04, 0.25]$	null
Emergent misalignment	0.51 ($n=23$)	-0.21	-0.07	$[-0.24, 0.16]$	null
Marker (*)	n/a ($n=3$)	excluded (gate fail; panel overlap $4 < 8$)			
Pooled ($n=399$)	n/a	0.02	0.088	$[-0.017, 0.171]$	inconclusive

villain), below the 8 floor; with the centering persona held out, 3 bystanders remain.

4 Discussion and Limitations

What it means. The base-rate target and the leakage target come apart. Activation-space alignment to a behavior direction *is* a meaningful read of a persona’s dispositional behavior: it recovers the base rate for sycophancy ($\rho=0.68$) and emergent misalignment ($\rho=0.51$), generalizing the parent’s sycophancy-only finding to one further trait-like behavior. But it carries no information about *where an implant leaks* once the base prior is accounted for. The out-of-sample leakage read is null for every behavior; the only clearance is a partialling artifact for one behavior, it fails the incremental-fit and reliability gates, and it does not survive a $17\times$ larger pooled read. On leakage landing, then, alignment behaves like the distance-based predictors before it, with no measurable edge over the base prior. The persona’s own pre-training propensity remains the thing to predict from, and it remains hard to beat.

Limitations.

- **Power.** The per-behavior leakage reads have ≈ 17 –23 held-out personas ($\text{MDE-}\rho \approx 0.62$). The pooled read addresses this and still finds no beat, but small effects below the pooled MDE cannot be ruled out.
- **Weak, sign-flipping causal gate.** The refusal and EM directions pass the gate by ~ 1 judge point on a 0–100 scale and are non-monotonic across strengths; the marker direction fails outright. A weak gate means the geometric reads it validates are themselves fragile.

- **Layer mismatch (refusal).** The gate passes at L7 but the predictive cosine is read at L14, where the refusal direction steers negative. The refusal nulls may be a wrong-layer artifact rather than a true absence of signal.
- **Trained-shift directions fold the implant in.** EM and the realized marker direction are read off trained adapters, not pure base-model reads, so they are reported as secondary; they cannot enter the beat-the-prior set on equal footing with the base-model reads.
- **Reused base rate judged by Haiku.** The sycophancy base rate is inherited with its original Haiku judge rather than the project’s current Sonnet-4.5 default, a cross-judge inconsistency carried for join fidelity.
- **Marker untestable.** Three independent exclusions remove marker from the H3 set, so the leakage question is unanswered for the one programmatic behavior.

Confidence. MODERATE. The base-rate generalization (H1) is a clean positive on two behaviors; the leakage conclusion (H3) is a well-powered, pooled, multiply-gated null/inconclusive, though it rests on directions validated by a weak causal gate and on reused leakage panels.

References

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